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2)

3.15)

we know: $V_\pi(s) = E[G_t | S_t = s]$

and $G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$

now we define a constant $c \in \mathbb{R}$ and shift all rewards by c

$$R'_{t+1} = R_{t+1} + c \implies G'_t = \sum_{k=0}^{\infty} \gamma^k R'_{t+k+1} = \sum_{k=0}^{\infty} \gamma^k (R_{t+k+1} + c)$$

$$G'_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1} + \sum_{k=0}^{\infty} \gamma^k c = G_t + c \sum_{k=0}^{\infty} \gamma^k$$

we recognize that $\sum_{k=0}^{\infty} \gamma^k$ is a geometric series hence:

$$\sum_{k=0}^{\infty} \gamma^k = \frac{1}{1-\gamma} \implies G'_t = G_t + \frac{c}{1-\gamma}$$

so our new $V'(s) = E_\pi[G'_t | S_t = s] = E_\pi[G_t + \frac{c}{1-\gamma} | S_t = s]$

now since we can seperate. Hence:

$$V'_\pi(s) = E_\pi[G_t | S_t = s] + E_\pi[\frac{c}{1-\gamma} | S_t = s] = \boxed{V_\pi(s) + \frac{c}{1-\gamma}} \rightarrow V_c = \frac{c}{1-\gamma}$$

So every $V_\pi(s)$ is shifted by a constant value and adding c does not change the relative ordering (does not change policy comparison)

$$V'_\pi(s_1) - V'_\pi(s_2) = V_\pi(s_1) - V_\pi(s_2)$$